Geomorphological Experiments for Amelia Island

An interactive simulation of barrier-island formation,
Holocene accretion, and Pleistocene welding

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Abstract

Amelia Island is a barrier island on the northeastern coast of Florida, at the southern end of the Georgia Bight. It exhibits the classic "double island" structure of the Bight: an older Pleistocene core (Silver Bluff terrace, ∼125 ka) overlain on its seaward side by a Holocene wedge that welded onto the core ∼5,000 years ago, trapping the salt marsh now known as Egans Creek between them. Since welding, the Atlantic face has prograded seaward, preserving a set of nested beach ridges most clearly visible at Fort Clinch State Park on the island's north end. This document specifies a set of computational experiments—an interactive web simulation, paired with browser-driven verification—that reconstruct the island's coastline through time. We ground the simulation in published sea-level data for northeastern Florida [1], in the formation theories of de Beaumont, Gilbert, McGee, and Hoyt [4], and in the narrative provided in lectures by the Conserve Nassau organization [10, 11, 12].

Contents

1 Motivation

The question that motivated this work is direct: *how did Amelia Island grow into its present shape, and can that growth be made visible?*

The geomorphic story of the island is well established but is rarely shown as a process. Most published material is text-and-diagram. A barrier island, however, is a moving system: sea level oscillates by hundreds of feet over glacial cycles, shorelines migrate tens of miles, and a sandbar a few miles offshore approaches the mainland over millennia and finally welds onto it. These are kinematic phenomena that benefit from a kinematic representation.

We propose a browser-based simulation in which the user drags a slider through ∼125,000 years of coastline change and sees the island form. The simulation is anchored on real geospatial data (modern shoreline from OpenStreetMap, terrain signatures from USGS digital elevation models) and on real sea-level curves [1, 2, 3].

2 Geomorphic background

2.1 Barrier islands as a class of landform

A barrier island is a long, narrow accumulation of sand parallel to a mainland coast, separated from it by a lagoon, estuary, or marsh. Three classical theories have been offered for how such islands form:

- **De Beaumont** (1845): offshore submarine bars built by shoaling waves emerge above sea level. Flume experiments later showed bars cannot grow above sea level by this mechanism alone, so this theory is now considered a partial contributor.
- **Gilbert** (1885, "spit-accretion"): longshore drift builds a sandy spit attached to the mainland; storms breach the spit, isolating an island. This mechanism is observed in some modern systems.
- **McGee** (1890, "submergence"): rising sea level drowns mainland beach ridges and dune fields, isolating them as islands with a new lagoon behind. Hoyt (1967, [4]) supported this view with stratigraphic evidence— specifically, the absence of shoreface sediment below early barriers, which would be expected if the bars had emerged from below.

Modern process-based work [5] treats these as endmembers of a self-organizing system whose behavior depends on three controls: the rate of relative sea-level rise (RSLR), the supply of sand, and the back-barrier accommodation space.

2.2 Two regimes: transgressive and regressive

Once formed, a barrier can be in one of two regimes (Table 1).

Modeling [5] identifies a critical threshold near 2–5 mm/yr RSLR: above it, barriers are typically transgressive; below it, they can stabilize and prograde. This threshold is consequential for Amelia.

2.3 From sandbar to vegetated barrier

A barrier island begins as a bare sandbar—a submerged or intertidal accumulation of sediment where waves can no longer transport sand further offshore. The transition to a stable, above-sea-

Table 1: The two regimes of a barrier island, after [5].

Regime	Drivers	Signature
Transgressive ("rollover")	RSLR high, or sediment-starved; overwash moves sand landward across the island	Washover fans on lagoon side; landward-migrating shoreline; ridges not preserved
Regressive ("progradational")	Sediment supply exceeds accommodation; new sand builds seaward of older shoreline	Beach-ridge plain (the nested arcs at Fort Clinch); seaward-younging

level barrier requires vegetation. The sequence is roughly:

1. **Submerged bar**: sediment accumulating offshore, below mean low water.
2. **Emergent intertidal sandbar**: highest crests exposed at low tide, bare or sparsely colonized by salt-tolerant pioneers (*Uniola paniculata*, *Spartina patens*).
3. **Vegetated barrier**: dune grasses slow the wind, trap windblown sand, and build foredunes. The foredunes shelter back-dune shrubs, and the shrubs in turn shelter trees.
4. **Mature barrier with maritime forest**: oaks, palmettos, and a closed canopy on stabilized terrain.

The vegetation feedback is load-bearing for the system. Without plants, a bare sandbar cannot retain sand against wind erosion, and it disperses. The older Pleistocene core of Amelia exhibits the mature end of this sequence: 100,000 years of subaerial exposure has produced deep soils, rounded slopes, and a maritime oak hammock. The younger Holocene wedge, by contrast, shows steeper slopes, less mature soils, and a less stratified plant community—it has had only 5,000 years above sea level.

For our reconstruction, the implication is that the offshore Holocene sandbar between 8 ka and 5 ka was likely bare or sparsely vegetated. After welding, the established vegetation network of the older Pleistocene island could spread laterally onto the new wedge, accelerating its stabilization.

2.4 Pioneer-plant species and the binding mechanism

Hopf's lectures [14] identify the specific plant species responsible for binding the loose quartz sand of Amelia's dunes:

- **Sea oats** (*Uniola paniculata*) — the most critical pioneer species. Highly salt-tolerant, thrives when partially buried by windblown sand, and possesses a fibrous root network extending up to ∼30 feet deep. These roots physically bind the loose quartz grains, turning a passive sand pile into a stable foredune.
- **Bitter panicgrass** (*Panicum amarum*) — ground cover that traps sand at lower elevations and stabilizes the dune face.
- **Railroad vine** (*Ipomoea pes-caprae*) — a creeping vine that spreads across bare sand surfaces, slowing wind and trapping the next increment of windblown grains.

When this vegetation is destroyed—by foot traffic, vehicle crossings, or storm overwash—a bowl-shaped depression called a *blowout* forms. Blowouts are conduits: during the next storm surge, seawater penetrates the island through them, eroding back-dune areas that would otherwise have been protected.

2.5 Soil pedology: Pleistocene Spodosols vs. Holocene Entisols

The age difference between the Pleistocene core and the Holocene wedge is recorded not only in topography but also in *soil maturity*. The video report [14] synthesizes Hopf's discussion of the two distinct soil profiles:

- **Pleistocene core** ($\sim$125 ka): Mature **Spodosols**. After $\sim$100,000 years of subaerial exposure, organic acids from decomposing leaf litter have leached down through the upper sand layers, mobilizing iron and aluminum oxides. These re-precipitate at depth as a dark, cemented **Bh (spodic) horizon**—a coffee-grounds-colored band visible in soil pits in the Fernandina downtown and Plantation areas. The presence of this horizon is diagnostic of long subaerial exposure and reduced drainage; it is also responsible for the acidic pH and the live-oak/Southern-magnolia maritime hammock vegetation.
- **Holocene wedge** ($\sim$5 ka and younger): **Entisols**— unconsolidated, young, active sands with essentially no horizon development. Color is light tan to white (high quartz, modest shell content). The plant community is the early-successional dune complex described above: sea oats, panicgrass, sea grape, saw palmetto.

This pedological contrast is independent confirmation of the double-island age structure. Two adjacent surfaces with such different soils cannot have weathered together; they were exposed at different times. The Bh horizon is essentially a slow geologic clock that has been ticking on the Pleistocene side for $\sim$100 kyr longer than on the Holocene side.

2.6 The transgressive sandbar—what it looks like

The offshore Holocene sandbar that approaches Amelia between $\sim$8 ka and $\sim$5 ka deserves its own treatment, because it is geomorphically the most interesting (and visually the most distinctive) feature in the animation. It is a *proto-barrier island*: a long, narrow ridge of sand parallel to the existing shoreline, separated from the mainland by a developing lagoon, intermittently emergent at low tide.

Modern analogs. Many present-day barrier islands look exactly like Amelia's offshore sandbar did at 8–5 ka:

- **Outer Banks, North Carolina** ($\sim$320 km long, 0.5–3 km wide) — the textbook transgressive barrier, still migrating landward.
- **Padre Island, Texas** ($\sim$182 km long, 0.5–5 km wide) — the longest barrier island in the world; same long-narrow geometry.
- **Fire Island, New York** ($\sim$50 km long, $\sim$0.5 km wide) — another active transgressive barrier.
- **Cumberland Island, Georgia**, immediately north of Amelia — the same Georgia Bight "double island" as Amelia, just slightly older. The Holocene wedge welded onto its Pleistocene core in the same process described here.

Why long and narrow. The form arises from three balanced controls:

- **Length** is set by along-shore continuity of sediment supply. Longshore drift moves sand parallel to the coast and accumulates it into a continuous ridge wherever sediment supply is positive. For Amelia's ancestor bar, the supply came primarily from the St. Marys River and from sand reworked across Cumberland Sound.
- **Width** is constrained by the equilibrium between wave erosion on the seaward face and sand-supply deposition. During the active transgressive phase, this equilibrium width is typically 100–500 m; modern transgressive barriers stay at this scale until sea-level rise slows enough for them to stabilize and widen.
- **Curvature** mirrors the existing shoreface. The bar forms parallel to whatever coastline already exists, so its plan-view shape follows the curve of Amelia's modern Atlantic shore (which is, in turn, inherited from the Pleistocene core's geometry).

The landward migration. While sea level rose rapidly (rates above ∼5 mm/yr), the bar could not stabilize at any fixed offshore position. Storm overwash carried sand from the seaward face across the bar to the lagoon side, depositing it as washover fans. Over centuries, this net landward sand transport—combined with continued retreat of the equilibrium profile—caused the entire bar to "roll over" shoreward. Estimated migration rate during the slowing- RSLR mid-Holocene is approximately 1 km per millennium, consistent with Frank Hopf's reference statement that the bar "started out about four or five miles out from where we are today on the coastline" at ∼8 ka [11].

Visualization choices. In the simulation, the bar appears as a pale dashed polygon east of Amelia, in two discrete snapshots (8 ka and 6 ka). Several simplifications apply:

- **The dashed outline** signals that the bar is an inferred, migrating feature rather than a fixed landform with a known boundary.
- **Width** is rendered at ∼300 m for visibility; real transgressive bars were probably 50–200 m wide at this stage and widening as they slowed.
- **Only the emergent crest** is shown; in reality the bar has a submerged base extending 1–3 km further seaward at depth.
- **Migration** is rendered as two discrete positions rather than a continuous motion—a fidelity-vs-clarity tradeoff. The intermediate position is implicit; the user reading the slider can mentally interpolate.

After welding at ∼5 ka, the bar polygon disappears from the animation and the Holocene wedge polygon takes over as the bar's permanent successor on the welded island.

2.7 Beach ridges at Fort Clinch—what Hopf describes vs. what we infer

The progradational ridge complex at Fort Clinch State Park appears in the simulation as a sequence of nested arcs accreting eastward from ∼5 ka to the present. The ridges are real and visible on USGS topographic maps and satellite imagery, but the doc needs to be clear about which parts of the model derive from Hopf's lectures and which parts are inferred from the broader literature.

What Hopf describes explicitly. Hopf discusses the *post-jetty* ridges—the outermost subset of the ridge complex, formed between 1881 and today as a direct consequence of the St. Marys Entrance jetty construction. In *Dunes Pt 1* he says [11]:

> *"...up against the state park boundary, and you can see our Holocene dune over there against the park, nice high dune, protective. And then this series of dune ridges that grew up after they put in the jetty—and then in front of that is the rock revetment built by the Corps of Engineers after Hurricane Dora hit in 1964..."*

and in *Dunes Pt 2* [12]:

> *"...the sand that would go in like on the flood tide sort of got trapped there, and that's how that little piece of brand new land got formed on the north end of the island... and there's a bunch of little dune ridges up on the island there up on that section."*

Hopf attributes the post-jetty ridges specifically to flood-tide sand being trapped against the south jetty after 1881. He confirms their location at the north end of Amelia (Fort Clinch State Park) and distinguishes them from the older, higher Holocene dune inland.

What is inferred from the literature. Hopf does *not* enumerate the older, *pre-jetty* ridges individually. The seven older ridges in our model (ages 5 ka, 4 ka, 3 ka, 2 ka, 1 ka, 500 yr BP, and pre-jetty 1880 CE) are inferred from OSL-dated comparable Florida barrier systems:

- Rink and Forrest [8] dated Cape Canaveral and Merritt Island beach ridges with quartz OSL, deriving an average accretion rate of ∼80 years per ridge and ∼135 m of seaward land accretion per century.
- Forrest et al. [9] dated the St. Vincent Island strandplain (Florida Panhandle) with OSL, getting inter-ridge intervals of 78–148 years for the younger ridges.

Applied to Amelia's ∼5,000-year post-welding progradation, those rates predict 30–60 ridges *if* ridge formation were continuous. In practice, only ∼8–15 major arcs are preserved at Fort Clinch (storm-deposited foredunes, not annual ones, and many smaller ridges presumably eroded or amalgamated). Our model renders eight, consistent with the major-arc count visible in satellite imagery and the USGS topographic map referenced in Hopf's lectures.

Bottom line. The ridge complex's existence and overall pattern is Hopf's; the specific per-ridge ages are interpolated from comparable Florida systems. We label individual ridges by their inferred ages but acknowledge the dating is by analogy, not direct OSL on Amelia.

2.8 The Georgia Bight "double island" pattern

Along the Atlantic coast of Georgia and northeastern Florida, the Holocene barrier islands did not form on a virgin shelf. They formed offshore of an older set of Pleistocene barriers that had survived as elevated remnants. As sea level rose through the Holocene, the offshore Holocene bars migrated landward and *welded* onto the Pleistocene cores. Hoyt and Hails [4] identified six major Pleistocene shoreline complexes in coastal Georgia, each a former barrier-lagoon system, now overlain on its seaward side by a Holocene wedge. This pattern produces the double-island islands of the Bight: Hilton Head, Tybee, Wassaw, Ossabaw, St. Catherines, Sapelo, the St. Simons group, Jekyll, Cumberland, and Amelia at the southern end [6].

2.9 Amelia in Hayes' Georgia Bight classification

Hayes' comprehensive Georgia Bight chapter [6] divides the bight into six along-shore *compartments*, each with a distinctive shoreline character. Amelia Island sits in **Compartment V** (St. Simons Sound to the St. Johns River mouth), which Hayes calls the **"Combo-Drumstick Region."** The compartment contains just three large drumstick-shaped islands: **Jekyll, Cumberland, and Amelia**. Hayes writes [6]:

> *"Each of the islands has a core of Pleistocene barrier-island sand with welded Holocene beach ridges usually attached to the northern and southern ends."*

This is the exact structure the simulation renders: a Pleistocene core with Holocene wedge welded onto its eastern face, and the most prominent beach-ridge complex preserved at the *north* end (Fort Clinch State Park). Three further Hayes findings inform the simulation:

- **Mixed-energy "drumstick" morphology.** Drumstick-shaped islands are characteristic of mixed-energy, mesotidal coasts—those with tidal range 2–4 m and moderate wave energy. The drumstick form arises from updrift sediment trapping by ebb-tidal deltas, producing an asymmetric island with a broad northern bulge (the "thigh"). Amelia displays this geometry; the Fort Clinch ridge complex is its northern bulge.
- **Compartment composition.** Compartment V's outer shoreline is 68% regressive mixed-energy barrier islands (the Holocene beach-ridge sections), 19% eroding Pleistocene island core, and 13% harbor/sound entrances. The 19% *eroding* Pleistocene cores is a useful reality check: the Pleistocene face is not stable. Amelia's western (Amelia River) side is subject to slow erosion from estuarine and tidal currents.
- **Narrow back-barriers from the Ocala Uplift.** Hayes attributes the narrow back-barrier regions of Compartment V's islands to the tectonic influence of the **Ocala Uplift**, a regional structural high that shoals the inner shelf. This same uplift explains why Amelia's Pleistocene core sits relatively close to today's mainland (the marsh between Amelia and the mainland is narrow compared to, say, the broader Mississippi Sound behind the Mississippi–Alabama barriers).

2.10 Sediment supply: a coastal-plain ("blackwater") river

A subtle but important point in Hayes [6] is that the St. Marys River—Amelia's principal sand source from the north—is a *coastal-plain* or *blackwater* river, not a piedmont river. Its drainage basin lies entirely within the coastal plain (it does not reach the Appalachian foothills). Hayes notes that coastal-plain rivers "presently deliver only second-cycle, reworked sediments to the shoreline." Amelia's Holocene beach-ridge sand is therefore *recycled Pleistocene sand*, not fresh first-cycle sediment from the mountains—a detail worth keeping in mind when reasoning about long-term sediment budgets and the post-jetty sand-starvation problem that Frank Hopf describes (the jetty intercepts this already-limited recycled-sand supply, accelerating central-Amelia beach erosion until the modern Nassau County Shore Protection Project began artificially restoring it).

Mineralogical signature. Beyond the bulk quartz, Hopf [14] identifies the heavy-mineral fingerprint of Amelia's sand as a marker of its ultimate Appalachian provenance: **zircon**, **staurolite**, and **tourmaline**—all metamorphic accessory minerals diagnostic of high-grade-metamorphic source rocks in the Appalachian–Piedmont hinterland. These minerals were eroded out of the mountains over 10^8 years and survived multiple cycles of transport and deposition because they are mechanically and chemically very resistant. Even though the St. Marys delivers reworked sand today, the original provenance signature is preserved. Frank Hopf also notes a high **titanium-oxide** content (the dark heavy-mineral fraction visible as the "black sand" streaks on Amelia's beach at low tide) that was once considered for commercial mining—an extraction that fortunately did not proceed at Amelia (though it did at Ponte Vedra to the south, where the resulting beach erosion is still felt).

2.11 Florida marine terraces

The Pleistocene core of Amelia sits on the Silver Bluff terrace, formed during the Sangamonian interglacial (Marine Isotope Stage 5e) when sea level stood ∼6 m above present [7]. Florida exposes a sequence of older terraces stepping up in elevation and back in age: Pamlico, Talbot, Penholoway, and Wicomico. Only the Silver Bluff terrace is relevant to Amelia.

3 Sea-level history used in the simulation

The most directly relevant sea-level reconstruction for Amelia is Hawkes et al. [1], which produced 25 sea-level index points spanning 8.0 ka to 2.0 ka from salt-marsh foraminifera at sites in northeastern Florida. Three findings from that work shape the simulation:

1. Relative sea level rose ∼5.7 m in northeastern Florida over the last 8.0 ka.
2. There is **no Holocene highstand** at this latitude—the rise has been monotonic. This contrasts with the central US Atlantic coast and the Caribbean, which show mid-Holocene highstands.
3. The pattern reflects the collapse of the Laurentide Ice Sheet's proglacial forebulge, which produces ongoing subsidence at this latitude.

We extend the curve back to 125 ka by combining [1] with the Caribbean Holocene curve of Toscano and Macintyre [3], the Engelhart and Horton US Atlantic database [2] for the mid- Holocene tail, and Lambeck et al. for the Last Glacial Maximum lowstand. The resulting target curve is summarized in Table 2.

Table 2: Sea-level snapshots used by the simulation. Negative values indicate sea level below present. Sources are cited where each value is anchored.

Years BP	**RSL (m)**	**Phase / source**
125,000	+6	Eemian (MIS 5e) highstand; Silver Bluff terrace
80,000	−20	MIS 5a/4 transition
20,000	−120	Last Glacial Maximum
14,500	−90	onset of Meltwater Pulse 1A
13,500	−72	end of MWP-1A (∼18 m rise in 500 yr)
8,000	−5.7	Hawkes et al. 2016 anchor
5,000	−2.0	Welding event (slow RSLR, ∼1 mm/yr)
2,000	−0.5	Late Holocene progradation
150	−0.2	Pre-jetty (1880 CE)
0	0.0	Present (2026 CE)

4 Narrative arc of Amelia Island

The reconstruction proceeds in six phases.

Phase 0: Pleistocene core forms (∼125 ka). During the Sangamonian highstand, sea level stood ∼6 m above present and a barrier island formed on the Silver Bluff terrace at the location of today's western Amelia. Behind it: lagoon and salt marsh. The same system that produced neighboring Cumberland Island.

Phase 1: Lowstand exposure (125–20 ka). Sea level fell ∼125 m through the Wisconsinan glaciation. The Pleistocene Amelia stood as an inland hill, with the shoreline retreating to a Last Glacial Maximum position roughly 130 km east of today's coast, near the −120 m bathymetric contour on the continental shelf. During this ∼100,000 year exposure, the Pleistocene sands matured: deeper soils formed, slopes rounded, oak-hammock vegetation established.

Phase 2: Holocene transgression (20–8 ka). As the Laurentide and Eurasian ice sheets retreated, sea level rose rapidly, particularly during Meltwater Pulse 1A (14.7–13.5 ka, ∼36 mm/yr). RSLR this fast keeps barriers in the transgressive regime: ephemeral sandbars formed, drowned, and reformed at landward positions on the migrating shoreface. None were preserved in their original locations.

Phase 3: The approach (8–5 ka). By 8 ka, sea level was within ∼6 m of present and rising more slowly (∼2–3 mm/yr per [1]). A persistent Holocene sandbar formed roughly four miles east of the Pleistocene core. The shallow water between the sandbar and the older island became a protected lagoon. Mud—which will not settle out of suspension on its own—accumulated as oysters and clams filtered it from the water and deposited it as faecal pellets. Salt marsh built upward on this biogenic mud. The future Egans Creek wetland began to form.

Phase 4: Welding (∼5 ka). RSLR slowed below the stabilization threshold (∼2 mm/yr). The Holocene sandbar stopped retreating, grew vertically and laterally, and finally welded onto the Pleistocene island—first at the south end of Amelia, then propagating north. The Egans Creek marsh was sealed in between, becoming a relict back-barrier lagoon. This is the most visually distinctive event in the simulation.

Phase 5: Progradation (5 ka–1880 CE). With slow RSLR, abundant longshore sand from the St. Marys River to the north, and the welded island as a fixed anchor, the Atlantic face of Amelia began to build seaward. Successive shoreline positions are preserved as beach ridges, each separated by 80–150 years on comparable Florida systems [8, 9]. At Fort Clinch on the north end, the curving plan form of the island favored maximum accretion, producing the nested arc-shaped ridges visible from the air and on USGS topography.

Phase 6: Human modification (1880–present). The St. Marys Entrance jetty, completed in 1881 and extended afterward, intercepted the southbound longshore sand transport at the north end of Amelia. The result was an additional fillet of accretion—the "post-jetty dunes" visible on the seaward side of the historical shoreline at Fort Clinch State Park—superimposed on the natural pattern over 150 years rather than the natural rate of 80–150 years per ridge. The Nassau County Shore Protection Project (USACE Jacksonville District, with Olsen Associates) now manages periodic beach nourishment on the developed central portion of Amelia.

5 Simulation design

5.1 Goal

The simulation is a single-page web application. The user opens `index.html` in a browser. A Leaflet [15] map centered on Amelia Island shows the present-day coastline on an OpenStreetMap basemap. A horizontal slider, ranging from 125 ka BP to the present, is positioned below the map. Dragging the slider re-renders the map with the polygons appropriate to that time slice: the Pleistocene core (when above sea level), the Holocene wedge in its current state (absent, offshore as a migrating bar, welded, or with progradational ridges), and the salt marshes. A side panel displays a one-paragraph caption summarizing the era, paraphrased from the source lectures and the published literature.

5.2 Geometry

Geometry is built from three sources:

1. **Modern island coastline**: queried from OpenStreetMap via the Overpass API and exported to GeoJSON. This is real public-domain data.
2. **Pleistocene/Holocene boundary on the island**: traced once from the USGS 3DEP digital elevation model. The Pleistocene core has the rounded, soil- mature signature of an emerged terrace; the Holocene wedge has the corrugated ridge-and-swale signature of progradational accretion. The boundary approximately follows Egans Creek through the central island.
3. **Beach ridges at Fort Clinch**: digitized from satellite imagery and DEM hillshade. We digitize 8 major ridges, consistent with the major arcs visible in the field.

Pre-welding paleo-coastlines (12 ka, 8 ka, 5 ka) are reconstructed by offset from the modern shoreline using the migration distances cited in the source lectures (3–5 miles offshore at 12 ka, approaching at $\sim$1 mile per millennium). These are labeled in the UI as interpretive reconstructions, not measured paleoshorelines.

5.3 Cross-section view

A secondary panel renders the modern cross-section described by the source lectures (Plantation, Little Nana, and Big Nana profiles): a young Holocene dune on the Atlantic face, the Egans Creek marsh in the center, the older Pleistocene core to the west, and the back-barrier marsh on the mainland side. The panel animates the accretion of the cross-section through time, illustrating the "two piles of sand on a bed of mud" description Frank Hopf uses.

5.4 Architecture

- `index.html` — single page, no build step.
- `viewer.js` — loads snapshot GeoJSONs, interpolates between them as the slider moves, manages layer styling.

- `style.css` — Pleistocene shown in a muted earth-tone; Holocene sand in a pale ridge-and-swale pattern; salt marsh in green; sea in blue.
- `data/` — one GeoJSON per snapshot, plus `sea_level.json`.
- `scripts/` — Python utilities for data preparation and Selenium- based verification.

6 Verification with Selenium

All HTML web tests are written in Selenium [16], per project standard. The test suite (`tests/auto/test_viewer.p` launches a headless Chrome browser, loads the application from a local HTTP server, waits for the Leaflet map to fully initialize, and exercises the time slider through each defined snapshot. For each snapshot, the test:

1. Sets the slider to the snapshot's target year.
2. Waits for the layer transition to complete and tiles to load.
3. Captures a full-page screenshot to `docs/fig/snapshot_N.png`.
4. Asserts that the expected layers are present in the DOM (`.layer-pleistocene`, `.layer-holocene`, `.layer-marsh`, etc. as appropriate to the era).

The same captured screenshots are referenced by this document via `\includegraphics`. The verification suite therefore serves two roles: it tests the application, and it generates the figures embedded here. This document is rebuilt by running the test suite, then `pdflatex` on this file.

7 User guide

7.1 Launching the simulation

The simulation is a single static HTML file with no server-side dependencies. From the project root, launch the bundled `start` script:

```
./start
```

The script checks whether port 8765 is already in use (and stops a stale process if so), starts a local HTTP server, and opens `http://localhost:8765/index.html` in the default browser. To stop the server later:

```
./stop
```

7.2 The map view

On load, the application centers on Amelia Island at the present-day timestep. The OpenStreetMap basemap shows the modern coastline, with the simulation's geologic layers drawn on top: the Pleistocene core in muted earth tone, the Holocene wedge in pale sand, the Egans Creek marsh in green, the Atlantic and Amelia River in blue. Standard Leaflet pan and zoom controls are available; the geologic layers re-render at every zoom level.

7.3 The time slider

A horizontal slider below the map ranges from 125,000 years BP at the left to the present (2026 CE) at the right. Dragging the slider re-renders the geologic layers to reflect the coastline at that time. The slider is logarithmic in years BP, so most of its travel is in the Holocene, where the welding event and beach-ridge progradation occur.

A play button below the slider animates the simulation forward from the current position at one of three speeds (1×, 5×, 20×). A reset button returns the slider to 125 ka.

7.4 The era panel

To the right of the map, an information panel updates as the slider moves. It shows the current era name, the year BP, the relative sea level (in meters versus today), and a one-paragraph caption describing what is happening at that moment. Caption text is paraphrased from the source lectures [10, 11, 12] and the published literature cited above.

7.5 The cross-section panel

A toggle in the upper-right opens a secondary panel showing the modern cross-section of the island at the Plantation profile, after the cross-sections in [10, 11, 12]. From west to east this shows: the Amelia River, the mainland salt marsh, the Pleistocene core, the Egans Creek marsh, the Holocene wedge with its beach ridges, the modern foredune, and the Atlantic Ocean. As the time slider moves, the cross-section animates to show how each component built up. The cross-section caption identifies the "two piles of sand on a bed of mud" structure that Frank Hopf emphasizes.

8 Results

The figures below are captured directly by the Selenium suite as it exercises the viewer through each defined snapshot. Each figure shows the full application interface: the Leaflet map (left, top), the cross-section panel (left, bottom), and the era panel (right) with current year, relative sea level, and the narrative caption.

9 Reproducibility

The figures above are not hand-curated. They are produced by running the Selenium suite, which drives the viewer through every defined snapshot, asserts that the expected layer kinds are rendered at each timestep, and saves a screenshot per snapshot. To reproduce:

```
python3 -m venv venv
source venv/bin/activate
pip install selenium Pillow youtube-transcript-api yt-dlp
python scripts/fetch_osm.py
python scripts/build_snapshots.py
python tests/auto/test_viewer.py
pdflatex docs/geomorphological-experiments.tex
```

The Selenium suite exits with status 0 when all snapshots render correctly. Failures are reported as "FAIL" lines and logged to `tests/auto/test_viewer.log`. The figures in `docs/fig/` are overwritten each run, so this document always reflects the current state of the viewer.

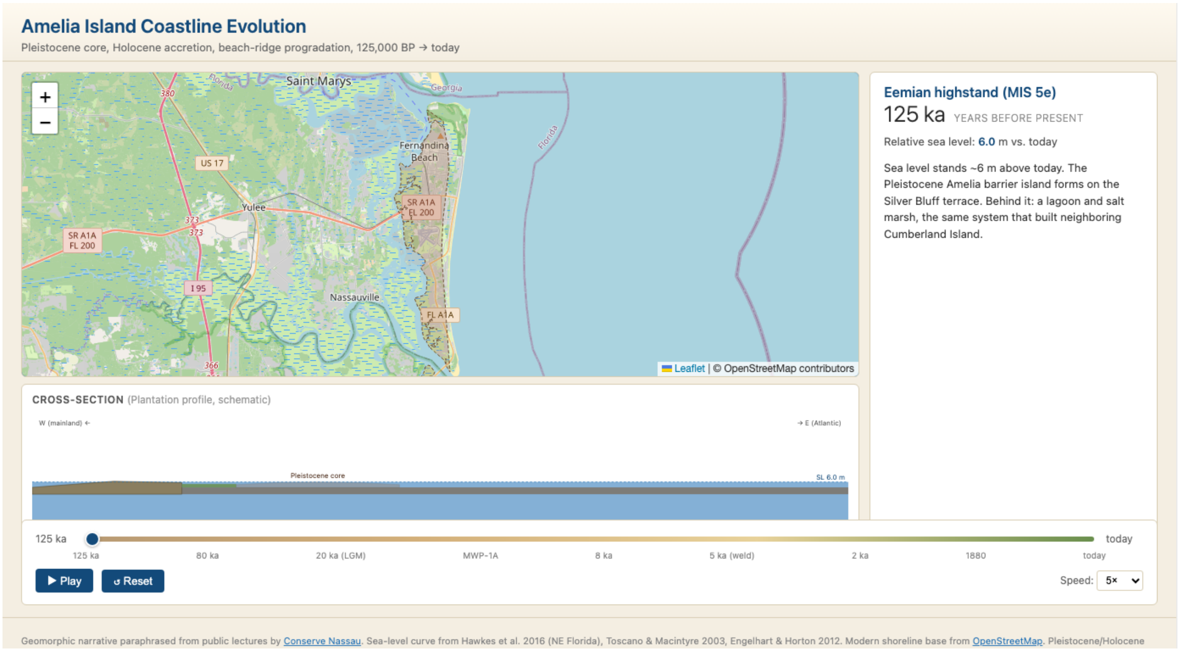

Figure 1: Eemian highstand, 125 ka. The Pleistocene core (Silver Bluff terrace) is forming under a sea level ∼6 m above today's. The cross- section shows the future-Amelia footprint as a single body above the dashed sea-level line; everything seaward of it will form later.

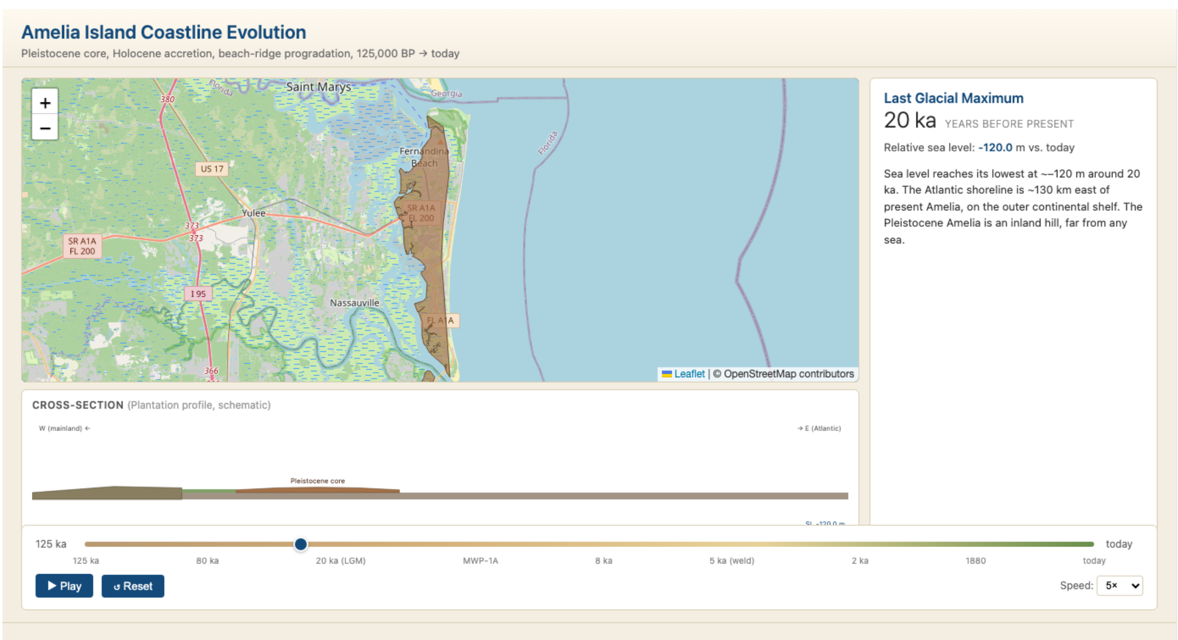

Figure 2: Last Glacial Maximum, 20 ka. Sea level −120 m. The red dashed hint line at the right edge of the map marks the LGM Atlantic shoreline ∼130 km east. The Pleistocene Amelia stands as an inland hill, far from any coast.

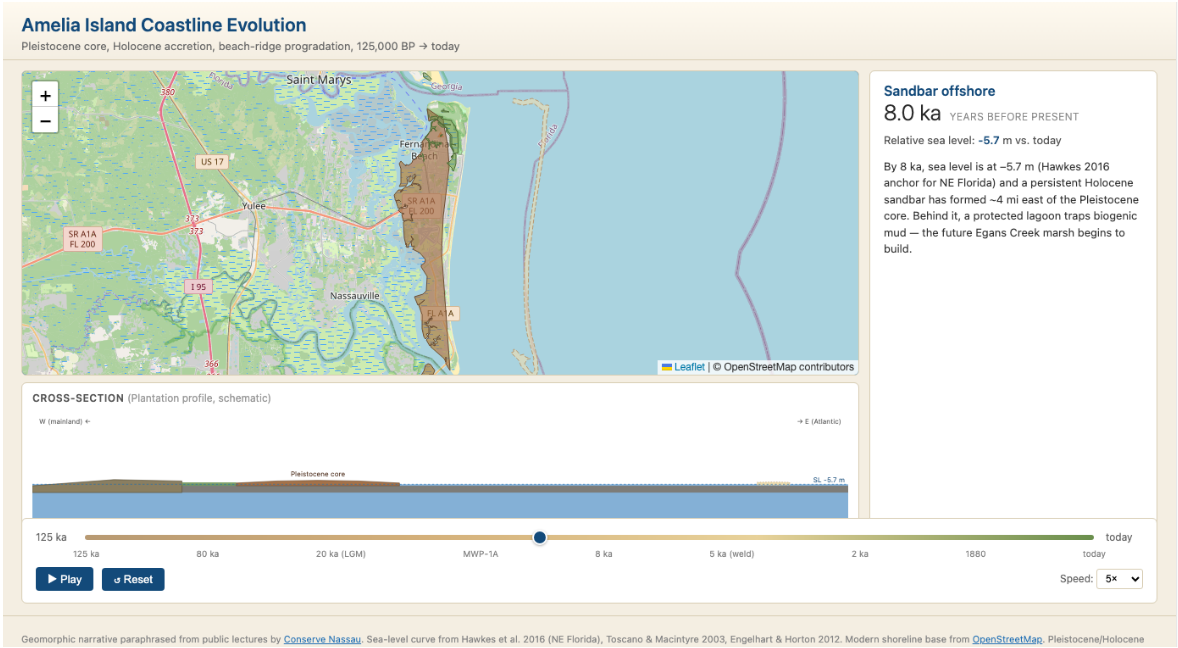

Figure 3: Holocene transgression, 8 ka. Sea level −5.7 m (Hawkes et al. 2016 anchor). A persistent Holocene sandbar (pale dashed polygon to the east of the island) has formed ∼4 mi east of the Pleistocene core. In the cross-section: the bar appears offshore (right), and the Egans Creek marsh begins to fill the lagoon between bar and core.

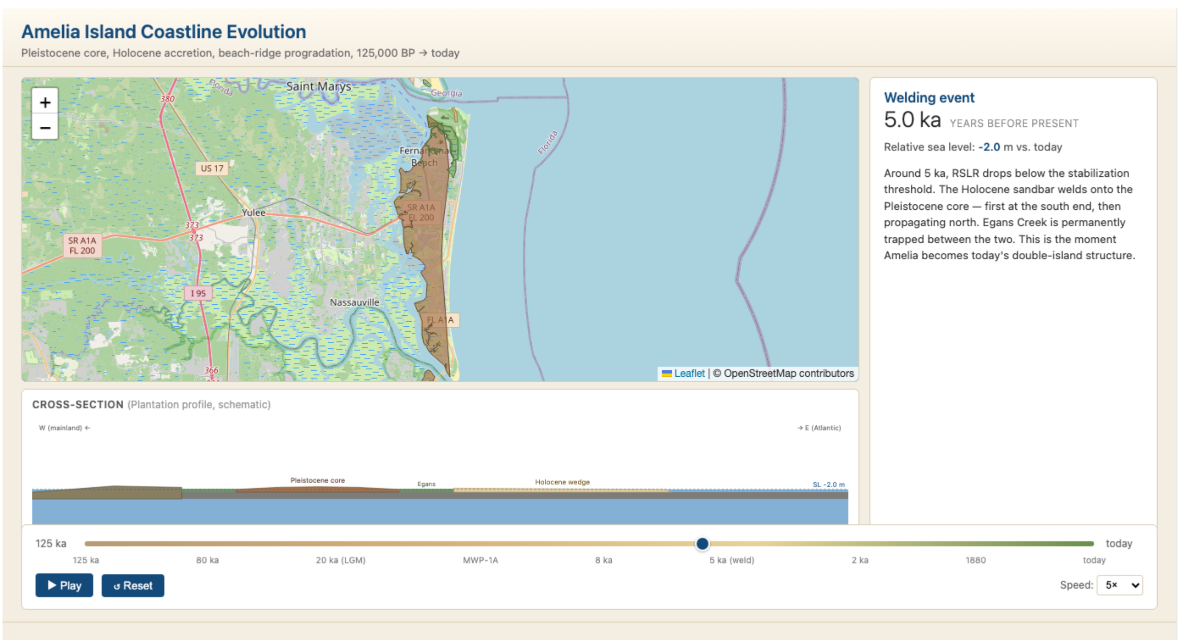

Figure 4: The welding event, 5 ka. The Holocene sandbar has welded onto the Pleistocene core. Egans Creek marsh is permanently trapped between them. This is the visual climax of the simulation: the moment Amelia becomes today's double-island.

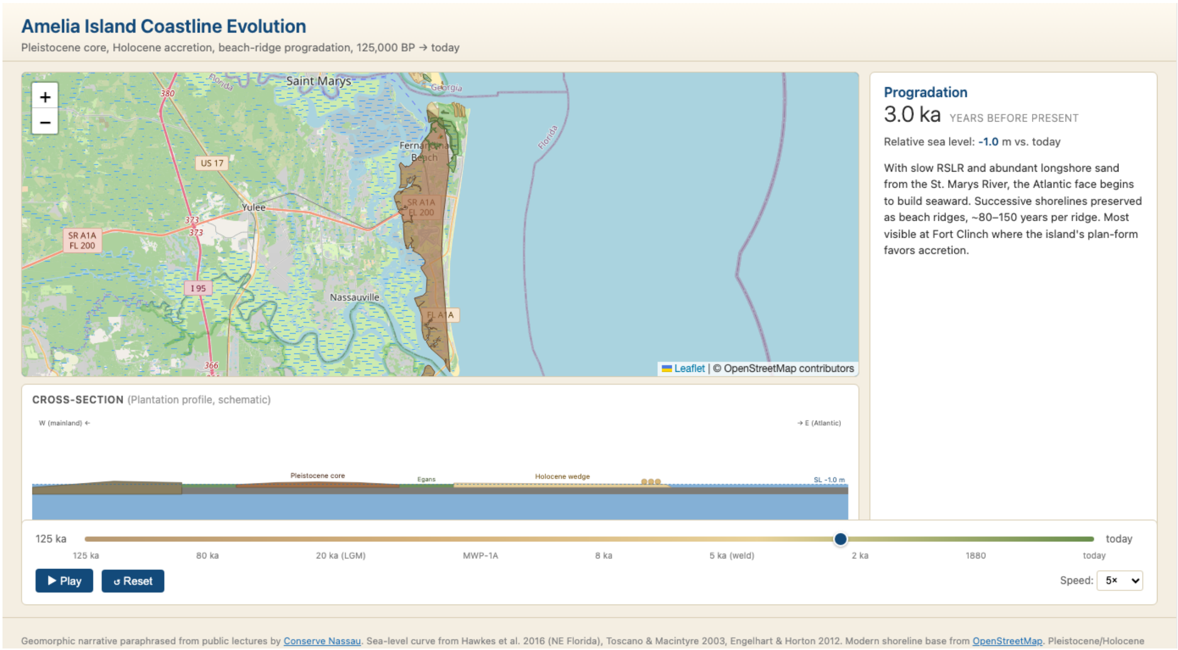

Figure 5: Mid-Holocene progradation, 3 ka. The Atlantic face begins building seaward. Beach ridges appear as small arcs near Fort Clinch (north end of the island). Sea level -1 m; sea-level rise has dropped well below the $\sim$2 mm/yr stabilization threshold.

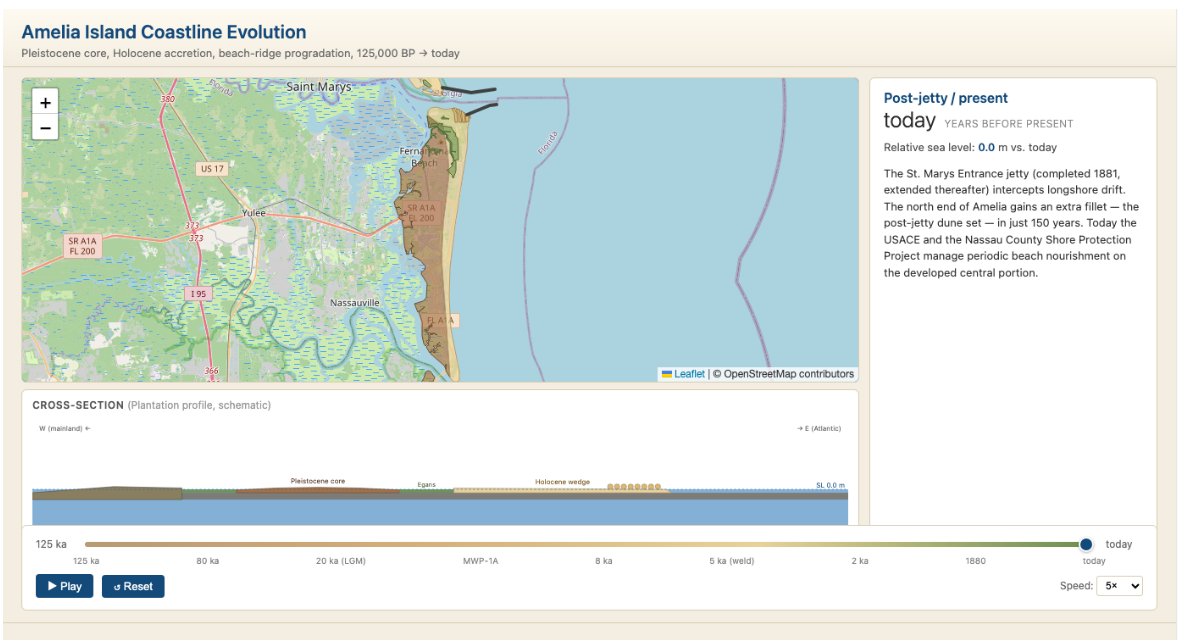

Figure 6: Present, 2026 CE. All eight modeled beach ridges are present at Fort Clinch, including the post-jetty fillet added since 1881. The cross-section shows the full “two piles of sand on a bed of mud” structure that Frank Hopf emphasizes: Pleistocene core on the west, Egans Creek marsh in the middle, Holocene wedge with beach ridges on the east.

10 Acknowledgements

The geomorphic narrative summarized here draws extensively from a series of public lectures given by **Dr. Frank Hopf, P.E.** (coastal geomorphologist and professional engineer), presented through the Conserve Nassau Florida organization [13]. Three of those lectures provided the narrative arc for the simulation:

- *Hopf: A Fragile Barrier Island* (64 min) [10] — the master overview of formation, sea-level history, and present-day fragility.
- *Dunes: Not Just a Pile of Sand — Part 1: Pangaea to Present* (28 min) [11] — the 400-million-year geologic context and specific spatial references for Amelia's Pleistocene and Holocene history.
- *Dunes: Not Just a Pile of Sand — Part 2: Our Holocene Dunes* (33 min) [12] — Holocene dune development and the post-jetty modifications since 1881.

A fourth source, the visual report [14], synthesizes Hopf's lectures via automated multimodal analysis of the video frames (in addition to the spoken-word transcripts). This source supplied the pedological terminology (Spodosols/Entisols, Bh horizon), the dune-plant Latin names (*Uniola paniculata*, *Panicum amarum*, *Ipomoea pes-caprae*), and the heavy-mineral provenance (zircon, staurolite, tourmaline)—specifics that the auto-captions of the spoken track did not reliably capture.

Transcripts were extracted programmatically from the YouTube auto-captions, then mined for dates, distances, ridge-set descriptions, and cross-section locations. Transcript material is paraphrased here under fair use for non-commercial educational illustration. Direct quotations are clearly marked. The original recordings remain the canonical source.

We also acknowledge the underlying peer-reviewed work that frames Amelia Island's geomorphology, particularly Hawkes et al. [1] for the NE Florida sea-level reconstruction, Hoyt and Hails [4] for the Georgia Bight stratigraphic framework, and Mariotti [5] for the modern process-based understanding of barrier-island stabilization.

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